

# DHRUV KUMAR

+91 6201675780 | kdhruv512@gmail.com | LinkedIn | GitHub | Bhopal, India

## ABOUT ME

Pre-final year Computer Science student specializing in AI and Machine Learning at VIT Bhopal. Winner of the InvokED I4EE Hackathon 2026 (2800+ participants). Experienced in building end-to-end intelligent systems — from fine-tuned NLP models to production-ready RAG pipelines — using modern full-stack and cloud technologies.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, C++, JavaScript, SQL (SQLite3), HTML, CSS

**Frameworks & Libraries:** Flask, Streamlit, React.js, React Native, Node.js, NumPy, Pandas, Matplotlib, Seaborn, OpenCV

**AI/ML:** Machine Learning, Deep Learning, NLP, Generative AI, RAG, BERT, TF-IDF, LLMs

**Databases & Tools:** MySQL, AWS, Docker, Git, GitHub, VS Code

## PROJECTS

**Guru Sikshan | React Native, Node.js, Flask, Gemini API, RAG**

**Feb 2026**

**1st place** at the Shiksha Lokam Innovation for Education Equity Hackathon 2026 (InvokED, HackerEarth) among **2800+ participants** — an AI-powered teacher support platform for underserved schools.

- Shipped a full MVP in a **2-week sprint**, building a React Native + Node.js app for structured teacher issue reporting and tracking.
- Engineered an AI-driven RAG pipeline that indexes **50+ pages** of training material in **under 10 seconds**, enabling instant retrieval.
- Reduced AI inference latency to under **3 seconds** for generating context-aware, personalized pedagogical feedback via Gemini API.

**Factualitytics | Python, BERT, LSTM, TF-IDF, Scikit-learn**

**2024**

An NLP-powered fake news detection system benchmarked across multiple deep learning and classical ML architectures.

- Trained and evaluated **4 models** (Random Forest, SVM, LSTM, XGBoost) on the LIAR dataset using TF-IDF and BERT embeddings.
- Fine-tuned BERT to achieve **76% accuracy** — a **50%+ improvement** over the SVM baseline — through systematic hyperparameter optimization.
- Designed a reproducible evaluation pipeline comparing statistical and neural approaches to multi-class misinformation classification.

**Hand Gesture Recognition System | Python, OpenCV, MediaPipe**

**2024**

A real-time, touchless computer control system using computer vision and hand landmark detection.

- Built gesture-based volume, zoom, and cursor controls achieving a stable **45 FPS** on standard consumer hardware.
- Designed for accessibility, with reliable detection at distances **up to 2 meters** and robustness to varying lighting conditions.

## EDUCATION

**B-Tech, Computer Science & Engineering | 2027**

**Class XII, CBSE | 2022**

**Class X, CBSE | 2020**

**Vellore Institute of Technology, Bhopal | CGPA: 8.17**

**Krishna Public School, Patna | 82%**

**Radiant International School, Patna | 94%**

## ACHIEVEMENTS

**1st Place — InvokED I4EE Hackathon 2026** (HackerEarth): Ranked **#1 out of 2800+ participants** for the Guru Sikshan platform addressing education equity in India.